Arman Soylu

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SUMMARY

M.Sc. Computer Science student at the Karlsruhe Institute of Technology (KIT), specialising in Machine Learning with a focus on NLP, Computer Vision and Reinforcement Learning. Two ML internships and end-to-end experience building, dockerising and deploying LLM-based systems (agentic RAG, MCP, FastAPI, Azure). Seeking a Werkstudent position in ML or software engineering — available up to 20 h/week during the semester and full-time during semester breaks.

WORK EXPERIENCE

Machine Learning Intern, Digiole – Helsinki, Finland (on-site & remote) *Sep 2024 – 2025*

- Developed an intelligent market-search and outreach system: an AI-powered pipeline that automates company research and personalised sales e-mails using web scraping and LLMs.
- Applied NLP techniques for spam detection in the automated e-mail pipeline.
- Deployed on AWS with automated e-mail generation and response tracking.

Technologies: Python, LLMs, NLP libraries, AWS, web scraping

Machine Learning Intern, Ditas – Niğde, Turkey *Feb 2024 – Jul 2024*

- Built a spam-mail detection desktop application using XGBoost.
- Implemented a real-time object detection and classification system.
- Designed DIAS (Dynamic Interview Automation System) to streamline HR document screening — later extended into the standalone project below.

Technologies: TensorFlow, LangChain, XGBoost, EasyOCR, Tesseract

PROJECTS

Portfolio Website & Agentic Research Assistant armansoylu.me

Full-stack personal site with a built-in AI research agent: React frontend and FastAPI backend implementing an agentic RAG architecture with MCP tool orchestration and SSE streaming. Fully dockerised and deployed on Azure Container Apps (GitHub Container Registry, Cloudflare DNS). *Stack: Python, FastAPI, React, Gemini API, Docker, Azure, MCP.*

Dynamic Interview Automation System (DIAS) GitHub

AI-powered system that converts static forms into adaptive LLM-driven interviews; reduced HR screening time by ∼50 %. *Stack: Python, Groq API, Llama 3, NLP.*

Transformer from Scratch GitHub

Implemented a complete encoder–decoder Transformer in pure PyTorch — multi-head attention, positional encoding, padding and causal masking — and trained it end-to-end. *Stack: Python, PyTorch.*

Multi-Model Image Classification System

Trained and compared EfficientNet, Xception, ResNet and a custom CNN for multi-class classification; interactive Qt UI with ensemble voting and configurable training parameters. *Stack: Python, TensorFlow/Keras, Qt Designer.*

EDUCATION

2025 – present	**M.Sc. Computer Science**, Karlsruhe Institute of Technology (KIT), Germany
2019 – 2024	**B.Sc. Computer Engineering**, Kastamonu University, Turkey
2021 – 2022	Exchange semester, PWSTE w Jarosławiu, Poland

SKILLS

AI / ML	PyTorch, TensorFlow, Keras, Hugging Face, scikit-learn
LLMs & Agents	Agentic RAG, MCP, LangChain, LLM fine-tuning (LoRA), Gemini & Groq APIs
NLP	Transformers, spaCy, NLTK
Computer Vision	OpenCV, YOLO, EfficientNet, ResNet
RL	OpenAI Gym, Stable-Baselines3
Dev & Cloud	Python, FastAPI, React, Docker, Azure Container Apps, AWS, Git

LANGUAGES

English (C1) • German (A2, ongoing course) • Turkish (native)